Teacher Wu Kuan | See It at a Glance #2 — $(a + b)^2 \neq a^2 + b^2$

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1. The trap question

Is $(a + b)^2 = a^2 + b^2$?

2. The method

- Square of side $a + b = a^2 + ab + ab + b^2$.
- $(a + b)^2 = a^2 + 2ab + b^2$.
- Quick test: $a = 2$, $b = 3 \rightarrow 25$ vs 13.

3. Parent check method

Have your child sketch the four pieces and label each area.

Practice

1. Expand $(x + 5)^2$.
2. With $a = 2$, $b = 3$, compare $(a + b)^2$ and $a^2 + b^2$.
3. Expand $(a + 2b)^2$.
4. Use $(30 + 1)^2$ to find 31^2.
5. $(x + y)^2 = 49$ and $xy = 10$. Find $x^2 + y^2$.

Answers

1. $x^2 + 10x + 25$
2. 25 vs 13
3. $a^2 + 4ab + 4b^2$
4. 961
5. 29

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